Enhancing Sustainability in Workflow Execution: Improving Job Caching and Meta-Scheduling in Galaxy

Nicola Soranzo[1], Marius van den Beek[2], Paul De Geest[3], Sanjay Kumar Srikakulam[4], Björn Grüning[4]

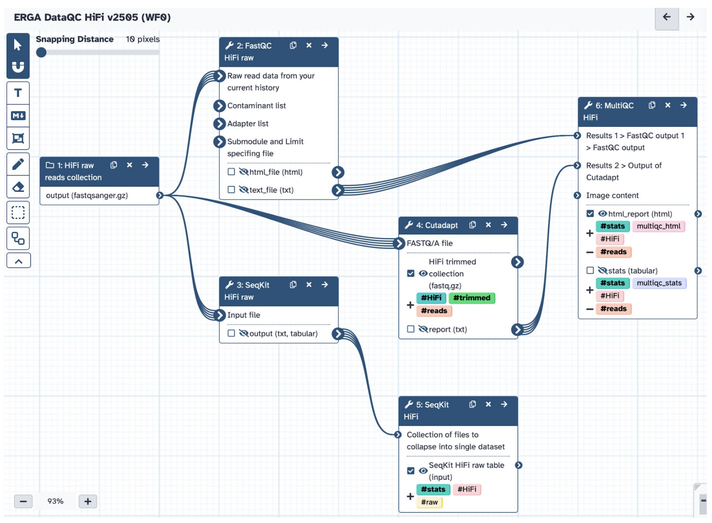
[1]Earlham Institute, Norwich, UK, [2]Sci-Scale, Belgium, [3]VIB, Belgium, [4]Albert-Ludwigs-Universität Freiburg, Germany

nicola.soranzo@earlham.ac.uk

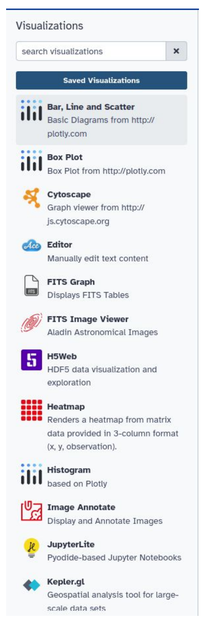
Galaxy - Web-based platform for FAIR data analysis

- Embeds any type of tool in an easy-to-use interface doi:10.1093/nar/gkae410
- **Open source** (Academic Free License 3.0 / MIT) with many free-to-use servers:
  - UseGalaxy.org, UseGalaxy.eu, UseGalaxy.org.au, UseGalaxy.fr, …
- **Cross-domain**: bioinformatics, chemistry, ecology, climate science, machine learning...
- >9,000 **tools** available on the Galaxy ToolShed https://toolshed.g2.bx.psu.edu/
- Makes your research **reproducible**: Galaxy keeps track of all analysis details
- **Workflows** to re-run or share an analysis
- Widely used (>13,000 publications on https://www.zotero.org/groups/1732893/galaxy)

Galaxy https://galaxyproject.org/



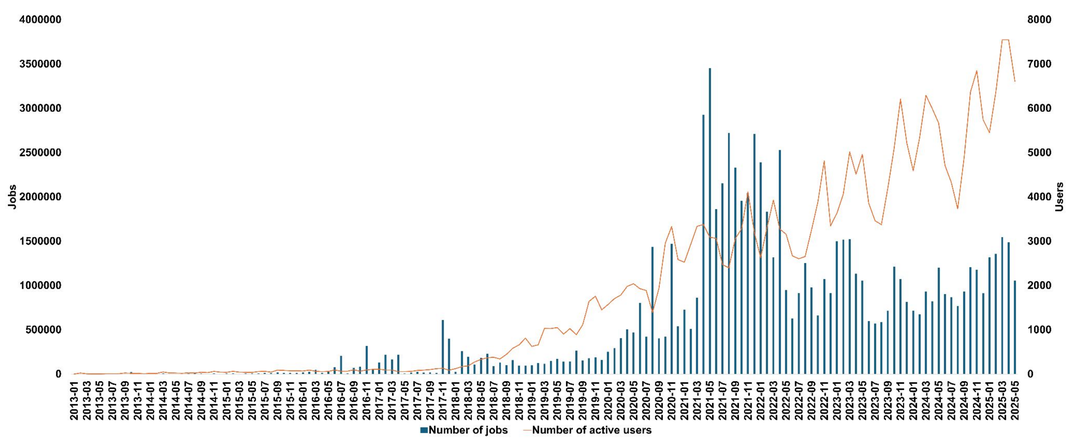
A Galaxy workflow from https://workflowhub.eu/workflows/602



>30 embedded visualizations

Environmental impact of computational research

- The ICT sector accounts for 1.8-2.8% of global greenhouse gas emissions (more than the aviation industry). While computational research represents only a small fraction of this, its emissions are already significant and expected to grow. doi:10.1038/s43588-023-00506-2
- Carbon emissions from computing vary widely depending on hardware, runtime, location (carbon intensity of electricity), and memory usage.
- Tools like Green Algorithms, CarbonTracker, CodeCarbon, and Cloud Carbon Footprint are emerging to help researchers estimate and track the carbon emissions of their computational analyses.
- Workflow systems like Galaxy are well placed to influence the environmental impact of 100,000s of researchers



UseGalaxy.eu usage data

ELIXIR-STEERS

- **ELIXIR** (https://elixir-europe.org/) is a European intergovernmental organisation that brings together scientists from >240 research institutes spread over 21 Member countries, 3 Observers and EMBL-EBI
- ELIXIR-STEERS is a €4M Horizon Europe project led by ELIXIR, involving 38 research institutes from 25 countries across Europe.
  - Objective 1: Create a toolkit for robust, reproducible, and **green software and workflows**.
  - WP 3 - Infrastructure services to enable adoption and deployment of software best practices
    - Task 3.4 - Integrating optimisation criteria for environmental impact in commonly used workflow management systems:
      - Make researchers **aware of the environmental impact** of an analysis
      - **Avoid unnecessary computations** in Galaxy by using a job cache
      - **Environmentally-friendly job scheduling** (in collaboration with **EuroScienceGateway**)
      - Improve GA4GH Task Execution Service (TES) support in Nextflow

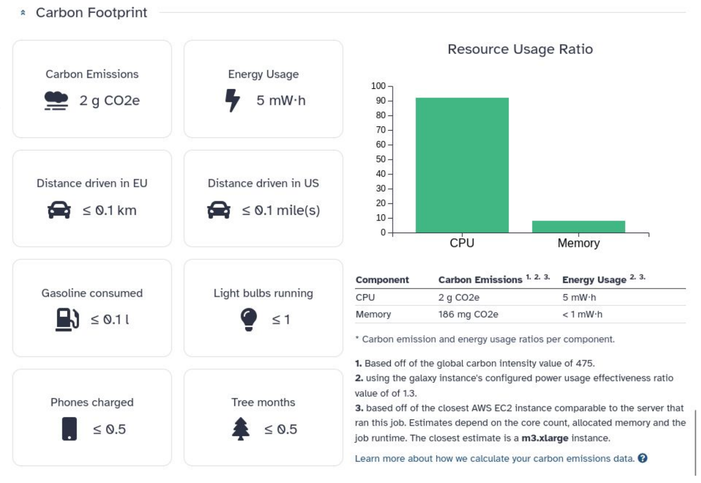
eosc | EuroScienceGateway

elixir STEERS

Raising environmental impact awareness

- Estimated carbon footprint shown in Galaxy job info
  - Calculations are based off of work done by the Green Algorithms project (https://www.green-algorithms.org/), and in particular their implementation of the "carbon footprint calculator". https://usegalaxy.eu/carbon_emissions_calculations
    - Estimated from: job runtime, memory allocated, CPU Thermal Design Power, CPU core count, number of cores allocated, and carbon intensity (amount of greenhouse gases emitted per unit of electricity, depends on the energy source used)

After running a tool in Galaxy, the carbon footprint of the job can be viewed by clicking the "i" (Dataset Details) icon in any of the tool output datasets.



Job cache to avoid unnecessary computations

- The Galaxy database stores data **provenance** information
  - Tool and version used, inputs, parameters
- Tools that produce deterministic outputs (large majority of Galaxy tools) should produce the same results if run again
- **Job cache**: before a tool is run, Galaxy can check if an identical job was computed previously
  - If the user has opted in to this feature (and the job outputs are still available), just link them into the current user's analysis history
    - Saving storage space as well
- Particularly effective for training courses (e.g. the yearly Galaxy Training Academy MOOC) where all trainees are following the same steps on the same data

Not computing at all is the greenest option! 🌳

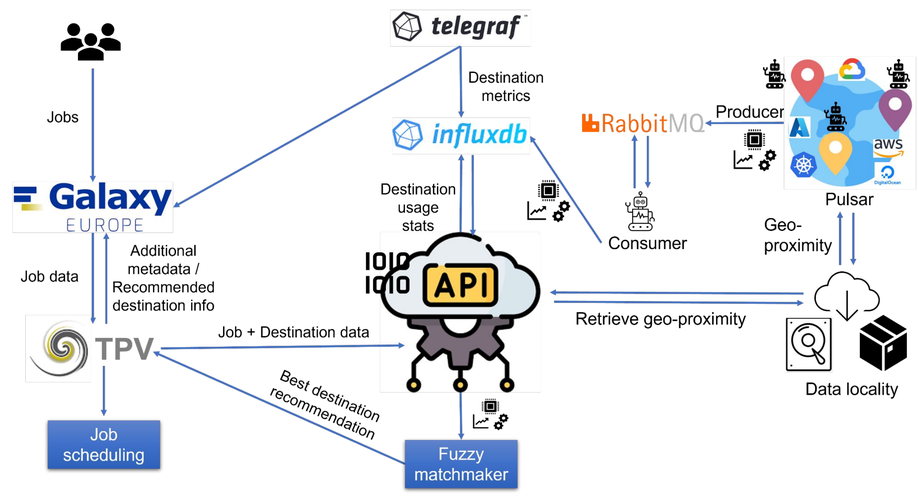
Initial implementation		Extended job cache
Required opt-in in the user preferences, and again when running a tool/workflow	→	Requires opt-in only when running a tool/workflow
Only allowed reuse of a user's own previous jobs	→	Allows reuse also of other users' jobs if they made their analysis **history public**
Inputs matched only if same dataset or copies	→	Inputs matched also if they have the **same hash**

The bulk of the job cache extension work was made at BioHackathon Europe 2024 (https://biohackathon-europe.org/), organised by ELIXIR.

TPV Broker for energy-aware job scheduling

- **TotalPerspectiveVortex** (TPV, https://total-perspective-vortex.readthedocs.io/) is a plugin for Galaxy that allocates resources (CPU cores, GPUs, memory) to jobs and maps them to appropriate compute destinations (HPC clusters, cloud) based on configured rules.
- **TPV Broker** introduces real-time, intelligent meta-scheduling and moves towards sustainable and energy-efficient computing.

TPV Broker architecture



Filter out destinations that cannot meet resource requirements → Calculate the geo-proximity between compute and data → Calculate the matching score for each destination → Sort destinations by matching score and geo-proximity → Schedule job to the best destination

Earlham Institute
Decoding Living Systems

Earlham Institute, Norwich Research Park, Norwich, Norfolk, NR4 7UZ, UK

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Biotechnology and Biological Sciences Research Council

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Acknowledgements

The ELIXIR-STEERS project receives funding from the European Union's Horizon Europe Programme under grant agreement number 101131096

eosc | EuroScienceGateway (https://galaxyproject.org/projects/esg/) is funded by the European Union programme Horizon Europe (HORIZON-INFRA-2021-EOSC-01-04) under grant agreement number 101057388 and by UK Research and Innovation (UKRI) under the UK government's Horizon Europe funding guarantee grant number 10038963.